

Detailed Explanation of an Improved YOLOv8n Road-Damage Detector

YOLO Fundamentals, SCConv, EMA, P2 Detection, and Multi-Scale Fusion

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Abstract

This document provides a detailed technical explanation of a custom YOLOv8n-based road-damage detector. It first covers the fundamentals of the YOLO family, then explains the YOLOv8n backbone components (Conv, C2f, SPPF), followed by detailed descriptions of the three main modifications: SCConv (Spatial and Channel Reconstruction Convolution), EMA (Efficient Multi-scale Attention), and the P2 detection head. It also covers the bidirectional P2-P3-P4-P5 neck, the four-scale detection head, loss functions, and the reasoning behind each design choice. The goal is to provide a complete reference suitable for a technical report or viva presentation.

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1 YOLO Fundamentals

1.1 What is YOLO?

YOLO (**You Only Look Once**) is a single-stage object detector. Unlike two-stage detectors (e.g., Faster R-CNN), YOLO predicts bounding boxes and class probabilities directly from the full image in one forward pass:

$$I \in \mathbb{R}^{H \times W \times 3} \longrightarrow \mathcal{D} = \{(x_i, y_i, w_i, h_i, c_i, p_i)\}_{i=1}^N.$$

Here:

- (x_i, y_i) = center coordinates of the bounding box,
- (w_i, h_i) = box width and height,
- c_i = class label,
- p_i = confidence score.

1.2 Single-Stage vs. Two-Stage Detection

Aspect	Two-Stage	Single-Stage (YOLO)
Pipeline	Region proposal + classification	Direct regression + classification
Speed	Slower	Very fast
Accuracy	High (especially small objects)	Competitive, improving
Typical models	Faster R-CNN, Mask R-CNN	YOLO, SSD, RetinaNet

Table 1: Single-stage vs. two-stage detectors.

1.3 Anchor-Free Detection in YOLOv8

YOLOv8 is **anchor-free**. Instead of predicting offsets from predefined anchor boxes, it predicts distances from each grid cell to the four sides of the object:

$$(t_x, t_y, t_w, t_h) \longrightarrow (\ell, t, r, b),$$

where ℓ, t, r, b are the left, top, right, and bottom distances from the grid anchor point. This:

- removes the need for anchor-box tuning,
- simplifies the prediction head,
- improves generalization across datasets.

1.4 Multi-Scale Detection

YOLO detects objects at multiple feature pyramid levels:

$$\text{P3 (stride 8), P4 (stride 16), P5 (stride 32).}$$

Small objects are detected at P3, medium at P4, large at P5. This work adds **P2 (stride 4)** for very small road damages.

1.5 Loss Function

YOLOv8 uses a composite loss:

$$\mathcal{L} = \lambda_{\text{box}} \mathcal{L}_{\text{CIoU}} + \lambda_{\text{cls}} \mathcal{L}_{\text{BCE}} + \lambda_{\text{dfl}} \mathcal{L}_{\text{DFL}},$$

where:

- $\mathcal{L}_{\text{CIoU}}$ = Complete IoU loss for box regression,
- $\mathcal{L}_{\text{BCE}}$ = Binary Cross-Entropy for classification,
- $\mathcal{L}_{\text{DFL}}$ = Distribution Focal Loss for box refinement.

1.6 Confidence and NMS

Predictions are filtered by confidence, then **Non-Maximum Suppression (NMS)** removes redundant boxes:

$$\text{IoU}(B_i, B_j) = \frac{|B_i \cap B_j|}{|B_i \cup B_j|}.$$

Boxes with IoU above a threshold are suppressed, keeping the highest-confidence box.

2 YOLOv8n Backbone Components

2.1 Conv Block

The basic YOLOv8 convolution block is:

$$\text{Conv} = \text{Conv2d} \rightarrow \text{BatchNorm} \rightarrow \text{SiLU}.$$

SiLU (Sigmoid Linear Unit) is:

$$\text{SiLU}(x) = x \cdot \sigma(x) = \frac{x}{1 + e^{-x}}.$$

It is smooth and non-monotonic, which helps gradient flow during training.

2.2 C2f Block

C2f (**Cross Stage Partial with 2 convolutions and fusions**) is the core building block of YOLOv8. It splits the input into two branches:

- One branch passes through multiple bottleneck blocks.
- The other branch is a shortcut.

All branch outputs are concatenated:

$$\text{C2f}(X) = \text{Concat}(X_0, B_1(X_0), B_2(X_0), \dots, B_n(X_0)),$$

where B_i are bottleneck blocks. This design:

- improves gradient flow,
- reuses intermediate features,
- maintains computational efficiency.

2.3 SPPF (Spatial Pyramid Pooling Fast)

SPPF applies multiple max-pooling operations with different kernel sizes in sequence:

$$\text{SPPF}(X) = \text{Concat}(X, \text{MaxPool}_5(X), \text{MaxPool}_5^2(X), \text{MaxPool}_5^3(X)).$$

This captures multi-scale contextual information without significantly increasing parameters. In this work, SPPF is applied at P5 to capture large road damages such as large potholes and alligator cracks.

3 SCConv: Spatial and Channel Reconstruction Convolution

3.1 Motivation

Standard convolutions produce redundant feature maps. In road-damage detection, many background regions (road texture, shadows, lane markings) activate similar channels, wasting capacity. SCConv addresses this by:

- suppressing spatial redundancy via the **SRU**,
- reducing channel redundancy via the **CRU**.

3.2 SCConv Structure

$$X \rightarrow \text{Conv} \rightarrow \text{BN} \rightarrow \text{SiLU} \rightarrow \text{SRU} \rightarrow \text{CRU}.$$

3.3 Spatial Reconstruction Unit (SRU)

The SRU normalizes the feature map using GroupNorm:

$$x_{\text{norm}} = \text{GroupNorm}(X).$$

Then a channel-wise importance weight is computed:

$$w = \sigma(\text{mean}_{H,W}(x_{\text{norm}})),$$

where σ is the sigmoid function and the mean is taken over spatial dimensions.

A binary gate is formed:

$$g = \mathbb{I}[w \geq \tau], \quad \tau = 0.5.$$

Features are divided into two groups:

$$X_{\text{high}} = g \odot X, \quad X_{\text{low}} = (1 - g) \odot X.$$

The SRU output is:

$$X_{\text{SRU}} = X_{\text{high}} + \alpha \cdot X_{\text{low}}, \quad \alpha = 0.5.$$

This preserves high-information features and attenuates low-information ones.

3.4 Channel Reconstruction Unit (CRU)

The CRU first reduces channels:

$$s = \max\left(8, \frac{C}{2}\right),$$

$$Z = \text{Conv}_{1 \times 1}(X_{\text{SRU}}) \in \mathbb{R}^{B \times s \times H \times W}.$$

Then a lightweight transformation is applied:

$$F(X) = \text{Conv}_{1 \times 1}(\text{SiLU}(\text{BN}(\text{Conv}_{1 \times 1}(Z)))).$$

A residual connection adds the original input:

$$Y = F(X) + X.$$

3.5 Why SCConv Helps Road Damage Detection

Road-damage images contain many irrelevant regions. SCConv:

- emphasizes feature responses useful for identifying damage,
- reduces redundant channel representations,
- improves small-crack representation by preserving informative features.

4 EMA: Efficient Multi-scale Attention

4.1 Motivation

Road damages are highly directional:

- longitudinal cracks (mostly vertical),
- transverse cracks (mostly horizontal),
- alligator cracks (multi-directional).

EMA uses horizontal and vertical pooling to capture these directional contexts.

4.2 EMA Structure

Let

$$X \in \mathbb{R}^{B \times C \times H \times W}.$$

Channels are divided into groups:

$$\text{groups} = \min(8, C).$$

4.3 Horizontal and Vertical Pooling

$$X_h = \text{AdaptiveAvgPool2d}((None, 1))(X), \quad X_w = \text{AdaptiveAvgPool2d}((1, None))(X).$$

These capture horizontal and vertical contextual information respectively.

4.4 Attention Generation

$$A = \sigma(\text{Conv}_1(X_h) + X_w),$$

where Conv_1 is a 1×1 convolution and σ is sigmoid.

4.5 Local Feature Extraction

A 3×3 convolution extracts local features:

$$L = \text{Conv}_{3 \times 3}(X).$$

The attention map weights the local features:

$$Y = L \odot A,$$

where $\odot$ denotes element-wise multiplication.

Finally, GroupNorm is applied and the input is added residually:

$$Y_{\text{EMA}} = \text{GN}(L \odot A) + X.$$

4.6 Why EMA Helps

- The residual connection preserves original information.
- The horizontal/vertical pooling emphasizes direction-aware features.
- Group processing reduces computation cost.

5 P2 Detection Head

5.1 Why Add P2?

Standard YOLO detects at P3, P4, P5. Small road damages (thin cracks, small potholes) can occupy very few pixels. After repeated downsampling to P3 (stride 8), fine spatial structure is lost.

P2 has stride 4 and resolution 160×160 for a 640×640 input:

$$\text{stride} = \frac{640}{160} = 4.$$

5.2 Comparison of Detection Scales

Feature	Resolution	Stride	Main Purpose
P2	160×160	4	Very small objects/details
P3	80×80	8	Small/medium objects
P4	40×40	16	Medium/large objects
P5	20×20	32	Large objects/context

Table 2: Detection scales for a 640×640 input.

5.3 Trade-offs of Adding P2

Advantages:

- Better preservation of fine details.
- Higher recall for small damages.
- Improved localization of thin cracks.

Disadvantages:

- Increased computation (feature map has 4x more spatial positions than P3).
- More memory usage.
- Potentially more false positives on textured road surfaces.

6 Bidirectional Multi-Scale Neck

6.1 Top-Down Path (Semantic Enrichment)

The top-down path propagates high-level semantics from P5 to shallower levels:

$$\begin{aligned} P5 &\xrightarrow{\text{Upsample}} \text{Concat}(P4) \xrightarrow{\text{C2f}} \xrightarrow{\text{EMA}} P4', \\ P4' &\xrightarrow{\text{Upsample}} \text{Concat}(P3) \xrightarrow{\text{C2f}} \xrightarrow{\text{EMA}} P3', \\ P3' &\xrightarrow{\text{Upsample}} \text{Concat}(P2) \xrightarrow{\text{C2f}} \xrightarrow{\text{EMA}} P2'. \end{aligned}$$

6.2 Bottom-Up Path (Detail Propagation)

The bottom-up path propagates fine-grained details from P2 to deeper layers:

$$\begin{aligned}
 P2' &\xrightarrow{\text{Conv}, s=2} \text{Concat}(P3') \xrightarrow{\text{C2f}} \xrightarrow{\text{EMA}} P3'', \\
 P3'' &\xrightarrow{\text{Conv}, s=2} \text{Concat}(P4') \xrightarrow{\text{C2f}} \xrightarrow{\text{EMA}} P4'', \\
 P4'' &\xrightarrow{\text{Conv}, s=2} \text{Concat}(P5) \xrightarrow{\text{C2f}} \xrightarrow{\text{EMA}} P5''.
 \end{aligned}$$

6.3 Why Bidirectional Fusion Works

- Top-down: adds high-level context to fine features.
- Bottom-up: adds fine-grained detail to high-level features.
- Combined: PAN-like fusion (Path Aggregation Network).

7 Complete Data Flow

```

INPUT: 640 x 640 x 3
|
Conv 3x3, s=2 -> 320 x 320
|
Conv 3x3, s=2 -> 160 x 160
|
C2f -> P2 (160 x 160 x 32) -----+
|
Conv 3x3, s=2 -> 80 x 80
C2f x2 -> SConv -> P3 (80 x 80 x 64)
|
Conv 3x3, s=2 -> 40 x 40
C2f x2 -> SConv -> P4 (40 x 40 x 128)
|
Conv 3x3, s=2 -> 20 x 20
C2f x2 -> SPPF -> P5 (20 x 20 x 256)
|
TOP-DOWN NECK
P5 -> P4 -> P3 -> P2 (with EMA)
|
BOTTOM-UP PATH
P2 -> P3 -> P4 -> P5 (with EMA)
|
DETECT: P2 / P3 / P4 / P5
|
Road Damage Predictions <-----+
|
Class, Box, Confidence

```


8 Component Contributions

Component	Location	Main Function	Benefit for Road Damage
C2f	Backbone/Neck	Feature extraction and gradient flow	Efficient feature learning
SCConv	P3, P4 backbone	Spatial/channel reconstruction	Reduces redundancy, improves informative features
SPPF	P5 backbone	Multi-scale spatial context	Captures larger damage
EMA	Neck	Spatial/channel attention	Focuses on important damage regions
P2	Detection neck/head	High-resolution feature detection	Better preservation of small cracks
Top-down path	Neck	Semantic to detail fusion	Adds context to fine features
Bottom-up path	Neck	Detail to semantic fusion	Propagates fine P2 info upward
P2/P3/P4/P5 Detect	Head	Multi-scale prediction	Detects damage across sizes

Table 3: Contributions of the main architectural components.

9 Why the Three Modifications Work Together

SCConv, EMA, and P2 solve different problems:

- **SCConv**: “What features should be represented more efficiently?”
- **EMA**: “Which parts of these features are important?”
- **P2**: “How can we preserve very small damage information?”

Together:

$$\boxed{\text{SCConv} + \text{EMA} + \text{P2}}$$

provides:

- better feature representation,
- better feature selection,
- better small-object information,

leading to improved multi-scale road-damage detection.

10 Training Details

10.1 Input Preprocessing

- Resize to 640×640 .
- Normalize pixel values to $[0, 1]$.
- Apply data augmentation: mosaic, mixup, HSV jitter, random flip.

10.2 Optimizer and Scheduler

- Optimizer: SGD with momentum, or AdamW.
- Initial LR: 10^{-2} for SGD, 10^{-3} for AdamW.
- Scheduler: cosine annealing or linear warm-up.

10.3 Evaluation Metrics

- Precision: $P = \frac{TP}{TP+FP}$.
- Recall: $R = \frac{TP}{TP+FN}$.
- F1: $F1 = \frac{2PR}{P+R}$.
- mAP@0.5: mean Average Precision at IoU 0.5.
- mAP@0.5:0.95: averaged over IoU thresholds 0.5 to 0.95.

11 Summary for Viva

A concise explanation:

Our model is based on YOLOv8n and enhances the backbone using SCConv for spatial and channel feature reconstruction, incorporates Efficient Multi-scale Attention in the neck to emphasize important spatial features during multi-scale fusion, and introduces a P2 detection branch at stride 4 to preserve high-resolution information for small road damages. A bidirectional feature-fusion pathway then propagates information between P2, P3, P4, and P5 before the final four-scale detection head.

12 Implementation Notes

- SCConv is inserted at P3 and P4.
- EMA is applied six times in the neck at channel sizes 128, 64, 32, 64, 128, and 256.
- The final detector uses P2, P3, P4, and P5.
- Input size: 640×640 .
- Number of road-damage classes: 5.